BS-PMN: Boundary-Aware and Self-Learning Prototype Matching Network for Few-Shot Remote Sensing Image Segmentation

Xuran Pan, Mengyao Chen, Lianru Gao, *Senior Member, IEEE*, Kexing Xu, and Xu Sun, *Member, IEEE*

***Abstract*—Addressing the critical challenges of blurred boundary delineation and high intraclass variance in few-shot remote sensing (RS) image segmentation, we propose a boundary-aware and self-learning prototype matching network (BS-PMN). While general few-shot learning methods provide solutions for label scarcity, their performance is often limited on RS data due to domain-specific challenges. Our work specifically addresses these distinct characteristics of RS scenes, including complex background interference, ambiguous object boundaries, and significant spectral variations. Our framework introduces three key innovations: 1) a multifeature frequency-aware prototype modeling (MFPM) module that integrates geometric neighborhood patterns, pixelwise spectral similarity, and frequency-aware components to generate discriminative prototypes through expectation–maximization (EM); 2) a boundary-aware prototype enhancement (BAPE) framework that incorporates structured boundary constraints through gated fusion and explicit boundary supervision to enhance spatial precision; and 3) a self-learning pseudolabel refinement (SPR) cycle that adapts support-derived knowledge to query-specific characteristics through pseudolabel guided refinement, effectively alleviating the domain gap between support and query samples. Experimental results on the iSAID-5^i and LoveDA-2^i benchmarks demonstrate that our method achieves highly competitive performance compared with existing approaches.**

***Index Terms*—Boundary-aware learning, few-shot segmentation (FSS), prototype learning, remote sensing (RS), self-learning refinement.**

I. Introduction

REMOTE sensing image segmentation is a cornerstone of Earth observation, enabling critical applications from urban planning to disaster response. While deep learning has revolutionized this field, its success is heavily reliant on abundant pixel-level annotations, a requirement that is often prohibitively expensive and time-consuming to meet for many specialized scenarios. Few-shot segmentation (FSS) has emerged as a promising paradigm to address this data scarcity issue by learning to segment novel object classes from only a handful of labeled support examples, primarily within a metalearning framework.

The evolution of FSS methodologies has been largely shaped by prototype learning, where features from support images are condensed into representative vectors (prototypes) to guide the segmentation of query images. However, the direct application of frameworks designed for natural images to the remote sensing (RS) domain has proven inadequate due to several domain-specific hurdles. First, RS imagery exhibits extreme intraclass appearance variance. Geo-objects within the same category (e.g., buildings) often present dramatically different spectral signatures and textures across different geographic locations and imaging sensors. This necessitates a robust prototype modeling strategy that can decouple stable structural features from complex spectral-structural variations, rather than relying on a single semantic vector. Second, there exists strong interclass similarity and complex background clutter. Many categories share similar visual patterns (e.g., roads and runways), leading to severe confusion between targets and backgrounds. To suppress such interference, an adaptive refinement mechanism is required to alleviate the support–query domain gap by leveraging query-specific information. Third, RS targets exhibit significant scale and shape variations. The irregular geometries of industrial facilities or airports are difficult to capture using coarse deep features. The inability of standard prototypes to delineate precise edges justifies the integration of explicit geometric constraints to provide fine-grained boundary priors, ensuring precise delineation.

As illustrated in Fig. 1, these limitations manifest as substantial performance degradation in practice. The significant intraclass appearance gap between the support mask (first row) and query mask (second row) highlights the challenge of extreme intraclass variance in RS imagery. Consequently, as shown in the third row, conventional prototype matching methods produce segmentation results with crude, blurred, and misaligned boundaries that fail to capture the precise geometric structures of the query objects. This pronounced boundary ambiguity severely limits their operational utility in real-world applications where precise delineation is crucial.

In response, recent research has pioneered adaptations specifically for RS. A primary trend involves moving beyond a single, coarse global prototype. For instance, Scale-Aware Detailed Matching (SDM) [1] and PCNet [2] propose extracting multiple prototypes to capture finer-grained details and intraclass diversity. Others, like R^2Net [3], employ dynam-

Received 25 September 2025; revised 17 January 2026 and 1 June 2026; accepted 9 July 2026. Date of publication 14 July 2026; date of current version 31 July 2026. *(Corresponding author: Lianru Gao.)*

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Digital Object Identifier 10.1109/TGRS.2026.3713109

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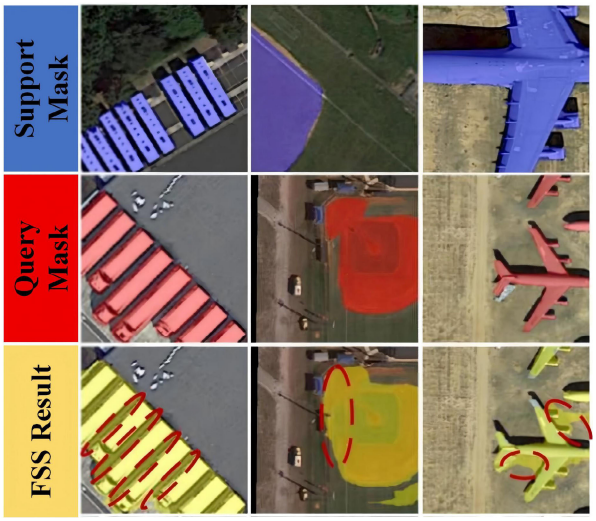


Fig. 1. Illustration of critical boundary errors in the conventional few-shot segmentation method.

ically updated global prototypes to suppress nontarget cues and prevent inaccurate activation. Another strategic direction focuses on leveraging the query image itself to mitigate the support–query domain gap. DMNet [4] introduces a "self-mining" perspective, emphasizing mining class-specific semantics from the query. Similarly, QPENet [5] proposes a query-guided prototype evolution mechanism, generating customized prototypes attuned to each specific query. Furthermore, recognizing the importance of boundary and spatial information, EPFNet [6] integrates global boundary information with prototypes to achieve more accurate matching.

Despite these significant advancements, critical limitations persist in addressing the unique RS hurdles mentioned above. First, many methods [1], [2], [3] still primarily operate on high-level semantic features, potentially overlooking discriminative low-level and frequency-domain cues (e.g., geometric texture and spectral similarity) that are crucial for distinguishing objects with high visual variance. Second, while some works [6] have incorporated boundary information, explicit and supervised modeling of precise boundary delineation remains underexplored, often leading to blurred segmentation masks. Finally, the process of generating robust query prototypes often relies on simple initial predictions [1], [2], [4], [5], [7], [8], which can be noisy and may propagate errors in the subsequent self-supporting cycle.

Motivated by these technical requirements, we propose a boundary-aware and self-learning prototype matching network (BS-PMN). Our core contribution is a comprehensive framework that explicitly maps the unique RS challenges to targeted architectural innovations.

1) *Multifeature Frequency-Aware Prototype Modeling:* To address the extreme intraclass variance in RS imagery, multifeature frequency-aware prototype modeling (MFPM) moves beyond conventional clustering in a single high-level semantic space. By integrating geometric neighborhood patterns, pixelwise spectral similarity, and frequency-enhanced components, the module introduces visual priors to guide the prototype estimation within the expectation–maximization (EM) process. This design ensures that the generated prototypes are not only semantically representative but also grounded in intrinsic object properties, effectively capturing essential features under diverse imaging conditions.
2) *Boundary-Aware Prototype Enhancement:* Targeting the challenge of blurred target edges and complex backgrounds in RS scenes, boundary-aware prototype enhancement (BAPE) transcends simple feature concatenation by employing an adaptive gated fusion mechanism to inject structured boundary priors into semantic prototypes. Unlike methods that treat boundary detection as a loosely coupled auxiliary task, this framework converts boundary supervision into a structural constraint within the prototype matching process. By directly rectifying spatial alignment ambiguities at the prototype level, BAPE significantly enhances the model's precision in delineating complex geo-object boundaries.
3) *Self-Learning Pseudolabel Refinement:* To mitigate the pronounced support–query domain gap in RS tasks, the self-learning pseudolabel refinement (SPR) cycle utilizes boundary-rectified prototypes as stable anchors for iterative prototype adaptation. Compared to generic self-mining methods prone to error accumulation in complex backgrounds, this cycle achieves robust adaptation through a filtered pseudolabel refinement process. This effectively suppresses the "mask leaking" phenomenon prevalent in low-contrast RS imagery and prevents error propagation, ensuring the generation of target-aware query prototypes.

In summary, our work represents a significant advancement in existing RS FSS methods in terms of multifeature fusion and boundary refinement, aiming to significantly improve the segmentation performance of RS targets with complex appearances and high-precision boundary requirements. We evaluate our method extensively on several few-shot RS segmentation tasks. Experimental results demonstrate that our approach achieves highly competitive performance, with significant gains in mean Intersection over Union (mIoU) and a notable reduction in boundary errors compared to strong baselines. The proposed framework not only addresses the inherent complexities of RS imagery but also establishes a new strong benchmark for the field.

II. Related Work

FSS aims to achieve pixel-level segmentation of novel categories with an extremely limited number of annotated samples. Its technical development builds upon fully supervised semantic segmentation models [9], [10], [11], [12], [13], [14] and few-shot learning paradigms [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]. Moreover, the unique challenges of RS images have spurred the emergence of numerous specialized methods [26], [27], [28]. This section systematically reviews the development trajectory of this field.

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A. Few-Shot Segmentation

The core idea of FSS has its roots in metric learning, particularly prototypical networks [20]. It computes a representative feature vector (prototype) for each category and performs segmentation by measuring the similarity between query pixels and prototypes. Shaban et al. [29] first proposed One-Shot Learning for Semantic Segmentation (OSLSM), a framework that applies metalearning to FSS with a dual-branch structure of a support set and a query set. Subsequently, Zhang et al. [30] proposed CANet, which achieves finer class-agnostic segmentation through an iterative optimization module and an attention mechanism, while Wang et al. [31] enhanced the feature consistency between the support set and the query set via the prototype alignment loss (PANet).

However, RS targets exhibit extremely high intraclass diversity, making it difficult for a single prototype to fully capture their complex features. To address this issue, researchers have proposed various multiprototype expansion schemes. Yang et al. [32] introduced prototype mixture models (PMMs), which employ multiple Gaussian components to model the category feature distribution. Liu et al. [33] and Li et al. [34] proposed a part-aware prototype network and an adaptive prototype learning and assignment strategy, respectively, to construct more expressive multiprototype representations in a dynamic manner. HSNet [35] further enhances the representational ability for targets with complex structures by building multiscale hierarchical prototypes. These methods have laid a solid foundation for subsequent research on RS FSS.

B. Few-Shot Segmentation in RS Images

Due to the extremely high intraclass variance, significant scale variations, complex background interference, and domain discrepancies between the support and query sets of RS images, FSS faces more severe challenges [1]. In recent years, researchers have proposed a large number of innovative methods to overcome these challenges, mainly focusing on the following directions.

1) *Multiscale and Multiprototype Matching:* To address the issue of scale variation in RS targets, Yao et al. [1] constructed iSAID-5^i, the first benchmark dataset for RS FSS, and proposed a scale-aware fine matching module. This module extracts multiple support prototypes to handle scale variability. Lang et al. [3] introduced dynamically updated global prototypes to provide auxiliary nontarget cues during matching, thereby preventing false activation caused by differences between support–query pairs. Mask-Guided Correlation Learning (MGCL) [36] introduced a mask-guided feature enhancement method, which effectively enhances the separability between foregrounds and low-contrast backgrounds. MS^2A^2Net [7] further proposed a multiscale self-attention aggregation network, which improves feature representational capacity through asymmetric momentum contrastive learning.
2) *Self-Support and Query-Aware Learning:* To alleviate the appearance discrepancy between support prototypes and query images, exploiting the intrinsic information of query images has become an important research direction. Fan et al. [37] pioneered the concept of self-support matching by generating query prototypes to align with query features. Bi et al. [4] proposed a "self-mining" perspective and designed the class public region mining (CPRM) and class-specific region mining (CSRM) modules, thereby shifting the paradigm from "learning from others" to "relying on oneself." Cong et al. [5] further introduced a query-guided prototype evolution mechanism, which generates customized prototypes through an iterative "support–query-support" process.

While these methods effectively exploit query-specific information, they predominantly rely on unfiltered initial predictions for self-mining. In complex RS scenes with high interclass similarity, such an approach is prone to error accumulation and "mask leaking." In contrast, our SPR cycle introduces a more stable paradigm by using boundary-rectified prototypes as anchors to guide the refinement, ensuring that the self-learning process is both target-aware and geometrically constrained.

3) *Boundary and Detail Optimization:* Accurate boundary delineation of RS targets is of great importance. Wu et al. [6] were the first to emphasize the significance of boundary information for RS targets and proposed a boundary-prototype fusion network, which fuses global boundary information with prototypes to achieve more accurate matching results. Lang et al. [2] leveraged base-class memory to enhance the understanding of novel-class prototypes through progressive parsing of support masks and commonality distillation. Jung et al. [38] introduced a boundary enhancement branch for building extraction, which significantly alleviated the problems of boundary adhesion and omission. Meanwhile, Wu et al. [39] proposed the conditional boundary loss (CBL), which improves boundary alignment through geometric constraints. However, existing works [6], [38], [39] often treat boundary information as an auxiliary supervision task or a simple feature fusion component. Our BAPE framework moves beyond this by establishing a gated interface that explicitly injects geometric priors into the semantic prototypes. This ensures that the boundary is not just a side-feature but a core constraint that rectifies the prototype matching process at the pixel level, a significant advancement over methods that lack such strong geometric guidance. In addition, Li et al. [40] developed the multiattention network (MANet), which achieved superior performance in preserving high-resolution details.

4) *Prototype Generation and Feature Representation Learning:* Based on prototype learning, numerous studies have explored more robust prototype generation methods. Shen et al. [41] introduced adaptive hyperprototype representation (HPR) and self-support matching to generate robust multiprototypes from both the support set and the query set. Deep meta-metric learning (DMML)-Net [42] conceptualized the segmentation task as a metric-based pixel classification problem and constructed a deep feature pyramid module. ProtoConViT [43] introduced a heterogeneous prototype distillation mechanism, fusing prototype information from Convolutional Neural Network (CNN) and Vision Transformer (ViT) to enhance the feature representation ability. Learnable

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TABLE I
TRAINING AND TESTING CLASS SPLITS OF THE iSAID-5^i, LoveDA-2^i, AND PASCAL-5^i DATASETS

Datasets	Training classes	Testing classes
iSAID-5^0	ground track field, bridge, large vehicle, small vehicle, helicopter, swimming pool, roundabout, soccer ball field, plane, harbor	ship, storage tank, baseball diamond, tennis court, basketball court
iSAID-5^1	ship, storage tank, baseball diamond, tennis court, basketball court, swimming pool, roundabout, soccer ball field, plane, harbor	ground track field, bridge, large vehicle, small vehicle, helicopter
iSAID-5^2	ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, bridge, large vehicle, small vehicle, helicopter	swimming pool, roundabout, soccer ball field, plane, harbor
LoveDA-2^0	water, barren, forest, agriculture	building, road
LoveDA-2^1	building, road, forest, agriculture	water, barren
LoveDA-2^2	building, road, water, barren	forest, agriculture
PASCAL-5^0	bus, car, cat, chair, cow, dining table, dog, horse, motorbike, person, potted plant, sheep, sofa, train, tv/monitor	aeroplane, bicycle, bird, boat, bottle
PASCAL-5^1	aeroplane, bicycle, bird, boat, bottle, dining table, dog, horse, motorbike, person, potted plant, sheep, sofa, train, tv/monitor	bus, car, cat, chair, cow
PASCAL-5^2	aeroplane, bicycle, bird, boat, bottle, bus, car, cat, chair, cow, potted plant, sheep, sofa, train, tv/monitor	dining table, dog, horse, motorbike, person
PASCAL-5^3	aeroplane, bicycle, bird, boat, bottle, bus, car, cat, chair, cow, dining table, dog, horse, motorbike, person	potted plant, sheep, sofa, train, tv/monitor

Prototype Few-Shot Segmentation (LPFS) [8] proposed a learnable prototype network, which eliminated interclass ambiguity and extracted high-level semantic prototypes based on a variational autoencoder. Despite the progress in robust prototype generation, most existing schemes operate in a high-level semantic space, potentially overlooking the visual properties of ground objects. Different from these methods, our MFPM reformulates the prototype clustering problem by integrating frequency components and structural–spectral similarity into an EM framework. This enables the generation of prototypes that are grounded in both semantic and visual priors, providing superior robustness against the extreme intraclass variance inherent in RS data.

Beyond the standard FSS settings discussed above, generalized FSS (GFSS) has recently emerged as a more complex and practical task extension [44], [45], [46]. GFSS requires the model to simultaneously recognize base and novel classes, shifting the focus toward balancing knowledge retention and new class adaptation. While our work, BS-PMN, operates within the standard FSS paradigm to address the fundamental hurdles of boundary precision and intraclass variance, the robust prototype modeling and geometric constraints we propose provide a critical foundation for these evolving generalized frameworks. Improving basic feature discriminability and boundary delineation remains a prerequisite for achieving superior performance in both standard and generalized FSS tasks.

III. METHOD

A. Overview

FSS in RS is significantly hindered by the critical challenges of blurred boundaries and high intraclass variance. To overcome these issues, we propose BS-PMN, an end-to-end trainable framework that seamlessly integrates boundary-aware prototyping and a self-learning refinement cycle, as depicted in Fig. 2.

The rationale behind this hybrid architecture lies in the sequential and targeted resolution of these fundamental challenges. Specifically, the MFPM module is designed to capture discriminative semantic prototypes even under extreme intraclass variance by integrating multisource features. Building upon this, the BAPE framework is integrated to rectify spatial ambiguities at geo-object boundaries by imposing explicit geometric constraints through a gated fusion mechanism. Finally, the SPR cycle is adopted to adaptively alleviate the support–query domain gap by generating target-specific prototypes. This synergistic combination ensures that the model preserves both semantic robustness and geometric precision. To provide a clear overview of the data processing workflow, the complete training procedure of BS-PMN is summarized in Algorithm 1.

Our method follows the standard episodic metalearning paradigm for FSS, aiming for the segmentation of novel classes with minimal annotations. The dataset is divided into a base class set C_{base} for metatraining and a novel class set C_{novel} for metatesting. The model operates episodically both during training and testing. Each episode ε is constructed to simulate a few-shot task, consisting of a support set $S = \{(I_s^k, M_s^k)\}_{k=1}^K$, containing K image-mask pairs for a target class, and a query set $Q = \{I_q\}$, containing unlabeled images to be segmented. The model's objective is to predict the query mask $\hat{M}_q$ for the class of interest present in the support set. The proposed pipeline processes each episode through three cohesive phases.

1) *Feature Encoding:* A shared-weight feature extractor $f_\theta(\cdot)$ (e.g., ResNet-50) first processes both the support images I_s^k and the query images I_q to generate their dense feature representations

$$F_s^k = f_\theta\left(I_s^k\right), F_q = f_\theta(I_q). \tag{1}$$

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Algorithm 1 Training Procedure of the BS-PMN Framework

Input: Training dataset D_{base}; Learning rate η; Maximum iterations T; Hyperparameter set Λ.

Output: Optimized model parameters $\theta*$.

Initialize network parameters θ;
for *iteration* = *1* to T do
Stage 1: Episodic Data Sampling
Sample a meta-training episode $\gamma = \{S, Q\}$ from D_{base}:$S = \{(I_s, M_s)\}$, Q=$\{I_q, M_q\}$;
Stage 2: Feature Encoding
Extract features F_s, F_q via backbone $f_\theta(\cdot)$ as in (1);
Generate support boundary mask B_s from M_s via the Canny operator;
Stage 3: Boundary-Aware Prototyping
Generate foreground prototypes P_{fore} and boundary prototypes P_{boun} as in (12);
Fuse prototypes to obtain P_{fused} =BAPE (P_{fore}, P_{boun}) using (20);
Compute initial prediction $\hat{M}_q^{\text{init}}$ using F_q and P_{fused} as in (4);
Stage 4: Self-learning Pseudo-label Refinement
Generate query pseudo-labels $\tilde{M}_q$ from $\hat{M}_q^{init}$;
Refine prototypes to obtain $P_{refined}$ using (5), and compute final prediction $\hat{M}_q^{\text{final}}$;
Stage 5: Optimization
Calculate total loss L_{total} using (21);
Update parameters: $\theta \leftarrow \theta - \eta \nabla L_{total}$;
end for
return $\theta^* = \theta$

2) *Boundary-Aware Prototyping and Initial Prediction:* This phase constitutes our core innovation for BAPE. The K support shots are spatially concatenated to form the composite support feature F_s, mask M_s, and its corresponding Canny-derived boundary mask B_s. The two masks guide the MFPM module Ψ to extract two independent sets of prototypes from the support features

$$P_{fore} = \Psi(F_s, M_s, I_s), \quad P_{boun} = \Psi(F_s, B_s, I_s) \tag{2}$$

where P_{fore} encodes the prototypical features of the object's interior, while P_{boun} encapsulates the prototypical features of its boundary transitions. This explicit disentanglement is the first crucial step toward robust representation.

We then synergize the disentangled representations. The semantic and boundary prototypes are fused via a boundary-aware fusion (BAF) module Φ_{fuse}, which employs cross-attention to compute relevance and a gating mechanism for adaptive blending

$$P_{fused} = \Phi_{fuse}(P_{fore}, P_{boun}). \tag{3}$$

The resulting fused prototype P_{fused} is a holistic representation that knows both what to segment and where its precise boundaries likely are. The initial prediction is generated by matching the prototype against F_q using a detail matching (DM) module, followed by refinement with an atrous spatial pyramid pooling (ASPP) module

$$\hat{M}_q^{\text{init}} = \sigma(\text{ASPP}(\text{DM}(P_{fused}, F_q))) \tag{4}$$

where σ denotes the softmax function. This prediction already benefits from explicit geometric guidance.

3) *Self-Learning Pseudolabel Refinement:* To alleviate the support–query domain gap and suppress the "mask-leaking" effect common in low-contrast RS scenes, we introduce a self-improving cycle. Unlike generic self-mining designs that rely on noisy initial predictions, our SPR initializes the cycle with boundary-rectified prototypes (P_{fused}) to ensure geometric consistency. The initial prediction for the query image $\hat{M}_q^{\text{init}}$ is regarded as a pseudo label $\tilde{M}_q$. This pseudolabel $\tilde{M}_q$ is then treated as a query-specific guidance pieced together with support mask M_s to extract the most relevant features from the query-support features via the MFPM module

$$P_{refined} = \Psi(\text{Piece}(F_q, F_s), \text{Piece}(M_q, M_s), \text{Piece}(I_q, I_s)) \tag{5}$$

where Piece $(\cdot, \cdot)$ denotes the operation of concatenating the two inputs in the spatial dimension. These self-generated, query-refined prototypes $P_{refined}$ are finally used to produce the final prediction $\hat{M}_q^{\text{final}}$, effectively completing the transition from external support to internal self-guidance.

Ultimately, this process effectively anchors the self-learning cycle within precise object boundaries, facilitating a stable domain transition from support-derived knowledge to query-specific characteristics. By seamlessly weaving together boundary-aware learning and iterative refinement, our framework progressively generates robust predictions that ensure superior performance and sharp boundary delineation, even in complex and low-contrast RS scenes.

B. Multifeature Frequency-Aware Prototype Modeling

Conventional prototype generation methods typically rely solely on high-level semantic features extracted by deep neural networks, which may overlook discriminative low-level cues that are crucial for RS imagery, such as textural patterns and spectral signatures. To overcome this limitation, we propose the MFPM module. Our approach generates discriminative prototypes via an EM algorithm applied to an enhanced feature representation, which integrates multisource information including geometric neighborhood patterns, pixelwise spectral similarity, and frequency-decomposed components. This integration yields highly discriminative and robust prototypes that are resilient to the extreme intraclass variance and complex backgrounds prevalent in RS imagery. The architecture of the MFPM module is illustrated in Fig. 3.

1) Frequency-Aware Decomposition and Fusion: To generate highly discriminative prototypes while mitigating the influence of high-frequency noise, we perform a spatial-based frequency-aware decomposition on the input feature

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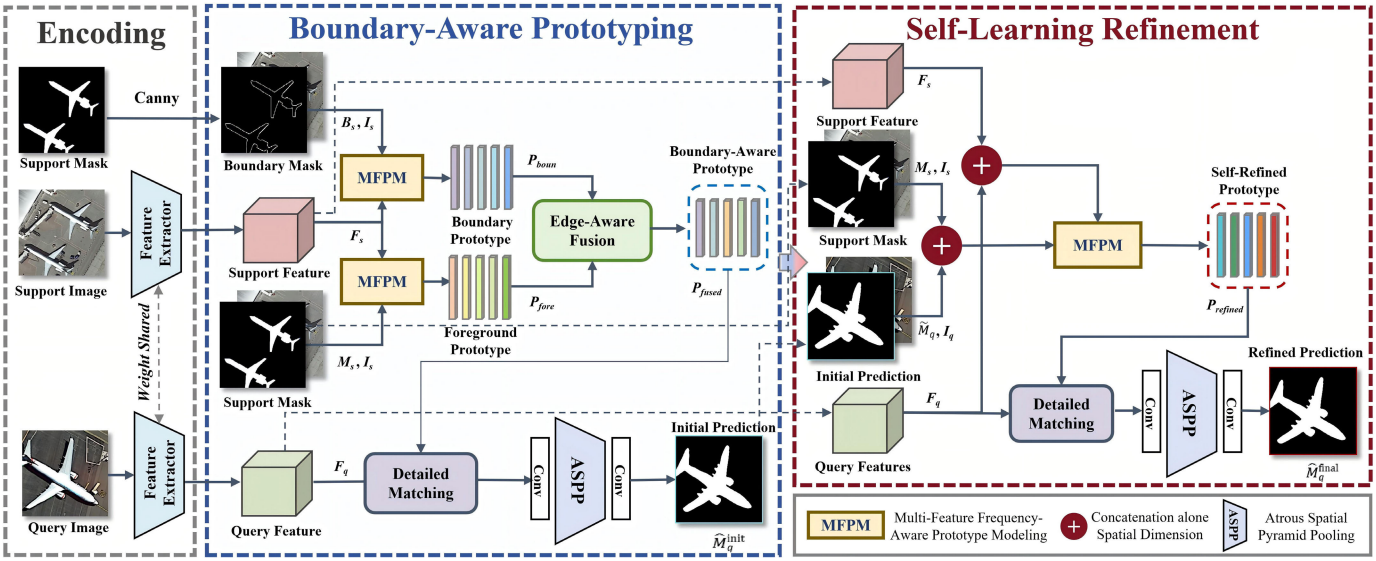


Fig. 2. Overall architecture of the proposed BS-PMN.

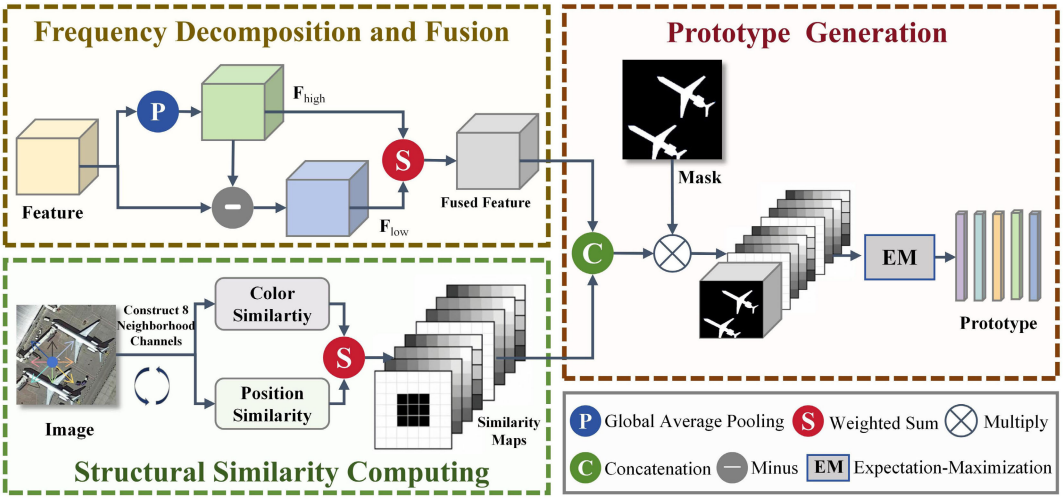


Fig. 3. Architecture of the MFPM module.

maps. Given a support feature map $F_s \in \mathbb{R}^{C\times H\times W}$ and its corresponding binary mask $M_s \in \{0,1\}^{1\times H\times W}$

$$F_{low} = \text{Interpolate}\left(\frac{\sum_{i=1}^{H}\sum_{j=1}^{W} F_s(:,i,j)\cdot M_s(i,j)}{\sum_{i=1}^{H}\sum_{j=1}^{W} M_s(i,j)+\epsilon}\right). \quad (6)$$

To ensure numerical stability and prevent division by zero, a small constant ϵ is included in the denominator.

The high-frequency component, encapsulating fine-grained details and potential noise, is then obtained by subtracting the low-frequency component from the original feature map

$$F_{high} = F_s - F_{low}. \quad (7)$$

Subsequently, these components are fused to produce a noise-suppressed yet detail-preserved feature representation. A weighting mechanism is employed for adaptive fusion

$$F_{out} = F_{low} + \alpha \cdot F_{high} \quad (8)$$

where α is a scaling parameter (empirically set to 0.5) controlling the contribution of high-frequency information. This process provides a more stable and informative feature foundation for subsequent prototype generation.

2) Structural-Spectral Similarity Computation: In addition to frequency information, we incorporate low-level structural and spectral cues to enhance feature discriminability. For an input red-green-blue (RGB) image $I \in \mathbb{R}^{3\times H\times W}$, we compute an eight-channel structural–spectral similarity map S. This is achieved by evaluating the multimodal similarity between each pixel and its neighbors along eight predefined directions, defined as D = {(±1, 0), (0, ±1), (±1, ±1)}.

For a displacement vector $d \in D$, the similarity S_d for each pixel location is calculated as a weighted combination of color similarity and spatial proximity

$$S_d(I) = \lambda_c \cdot \exp(-\gamma_c \cdot \|I - \text{Shift}(I,d)\|_2) - \lambda_p \cdot \exp(-\gamma_p \cdot \|P - \text{Shift}(P,d)\|_2) \quad (9)$$

where P denotes a normalized coordinate matrix encoding pixel positions, Shift (·, d) denotes the shift operation, and λ_c, λ_p, γ_c, and γ_p are balancing parameters. The final similarity

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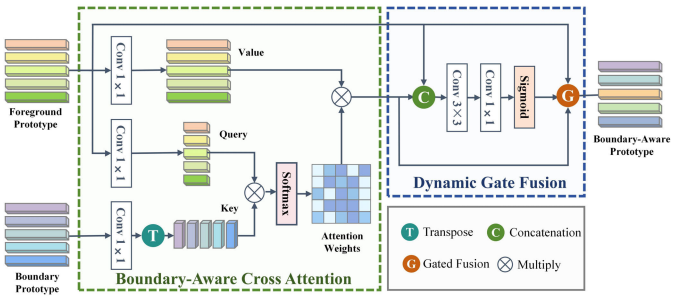


Fig. 4. Architecture of the BAF module.

map $S \in \mathbb{R}^{8\times H\times W}$ is formed by concatenating the similarity maps for all directions: $S = \oplus_{d\in D} S_d$.

This similarity map is then concatenated with the frequency-aware features F_{freq} and the original features F. The combined feature vector is passed through a 1×1 convolution layer for dimension reduction and fusion, yielding an enhanced feature representation $F_{enhanced}$

$$F_{enhanced} = \text{Conv}_{1\times 1}(\text{Concat}[F, F_{freq}, S]). \tag{10}$$

3) Prototype Clustering via EM Algorithm: The final step involves clustering the enhanced features $F_{enhanced}$ into a set of representative prototype vectors. We formulate this as a probability density estimation problem and employ the EM algorithm to generate the prototypes.

Let $X = \{x_1, x_2, \ldots, x_N\}$ represent the feature vectors within the masked region of $F_{enhanced}$, where N denotes the total number of foreground pixels. We aim to fit a Gaussian mixture model (GMM) with K components (prototypes). The EM algorithm proceeds as follows.

a) Initialization: The initial prototypes $\{\mu^{(0)}\}_{k=1}^{K}$ are either randomly sampled from the feature distribution or initialized using a predefined template.

b) E-step (Expectation): Calculate the posterior probability $\gamma_{nk}^{(t)}$ (responsibility) that features x_n belongs to the κth prototype

$$\gamma_{nk}^{(t)} = \frac{\exp\left(k \cdot x_n^{\mathrm{T}} \mu_k^{(t)}\right)}{\sum_{j=1}^{K} \exp\left(k \cdot x_n^{\mathrm{T}} \mu_j^{(t)}\right)} \tag{11}$$

where κ is a temperature parameter sharpening the distribution.

c) M-step (Maximization): Update the prototypes by computing the weighted mean of the features

$$\mu_k^{(t+1)} = \frac{\sum_{n=1}^{N} \gamma_{nk}^{(t)} x_n}{\sum_{n=1}^{N} \gamma_{nk}^{(t)}}. \tag{12}$$

The updated prototypes are then L_2-normalized. After T iterations, the EM process converges, yielding the generated prototypes $P = [\mu_1, \mu_2, .., \mu_K]$. This probabilistic clustering ensures that the prototypes effectively capture the underlying data distribution of the support set.

C. Boundary-Aware Prototype Enhancement

Although the MFPM module generates semantically meaningful prototypes, they may lack precise spatial boundary information, which is crucial for accurate segmentation. To address this limitation, we propose a BAF module. This module is designed to explicitly augment the semantic prototypes with structured boundary priors, ensuring that the guiding prototypes are not only class-discriminative but also spatially precise. The architecture of this module is depicted in Fig. 4.

Given the main semantic foreground prototypes $P_{fore} \in \mathbb{R}^{B\times K\times C}$ and the boundary prototypes $P_{boun} \in \mathbb{R}^{B\times K\times C}$, the fusion process is conducted as follows. First, the prototype vectors are reshaped into 2-D spatial maps P_{fore}^{map}, $P_{boun}^{map} \in \mathbb{R}^{B\times K\times C\times 1}$ to facilitate spatial reasoning. To capture the interplay between semantic and boundary information, a cross-attention mechanism models the relevance between the two prototype sets

$$Q = \text{Conv}_{1\times 1}\left(P_{fore}^{map}\right) \tag{13}$$

$$K = \text{Conv}_{1\times 1}\left(P_{boun}^{map}\right) \tag{14}$$

$$V = \text{Conv}_{1\times 1}\left(P_{fore}^{map}\right) \tag{15}$$

$$E = \text{Softmax}\left(\frac{Q \cdot K^{\mathrm{T}}}{\sqrt{d_k}}\right) \tag{16}$$

$$A = V \cdot E \tag{17}$$

where Q denotes query projection, K denotes the key projection, V denotes the value projection, and E is the attention energy map. The attended output A represents the boundary-refined semantic features. To adaptively fuse this with the original semantic prototypes, a gating mechanism is employed. The gate is generated by processing the concatenation of the original and attended features

$$G_{in} = \text{Concat}\left(P_{fore}^{map}, A\right) \tag{18}$$

$$G = \sigma(\text{Conv}_{1\times 1}(\text{ReLU}(\text{Conv}_{1\times 1}(G_{in})))) \tag{19}$$

where σ denotes the sigmoid function. The final fused, boundary-enhanced prototypes are obtained through a gated combination

$$P_{fused}^{map} = G \odot A + (1-G) \odot P_{fore}^{map}. \tag{20}$$

The resulting tensor P_{fused}^{map} is then squeezed back to the vector form $P_{fused} \in \mathbb{R}^{B\times K\times C}$. These prototypes now embody a synergistic representation that integrates both internal semantic homogeneity and boundary-specific characteristics, providing superior guidance for the subsequent segmentation of the query image.

D. Optimization Objective

To effectively train the proposed BS-PMN and enforce the constraints of its key components, including boundary-aware segmentation, self-learning cycle, and domain alignment, we design a composite loss function L_{total}. This loss function provides comprehensive supervision by integrating multiple objective terms, each targeting a specific aspect of the model's performance. The total loss is defined as follows:

$$L_{total} = \lambda_{init} L_{seg}^{init} + \lambda_{ref} L_{seg}^{ref} + \lambda_{boun} L_{boun} + \lambda_{domain} L_{domain} \tag{21}$$

where λ_* are balancing coefficients that control the importance of each term. The specific values used in our experiments are

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provided in the experimental section. Each loss component is detailed below.

The segmentation losses L_{seg}^{init} and L_{seg}^{ref} serve as the principal supervision, measuring the pixelwise error between the predictions and ground truth. We employ the standard cross-entropy loss for both the initial prediction $\hat{M}_q^{\text{init}}$, generated from boundary-enhanced support prototypes, and the refined prediction $\hat{M}_q^{\text{ref}}$, generated after the self-learning cycle

$$L_{seg} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{c=1}^{C} 1[y_{ij} = c] \cdot \log(\hat{p}_{ijc}) \tag{22}$$

where $\hat{p}_{ijc}$ denotes the predicted probability of class c at pixel (i, j), $1[\cdot]$ denotes the indicator function, and y_{ij} denotes the ground-truth label. The losses L_{seg}^{init} and L_{seg}^{ref} are computed using $\hat{M}_q^{\text{init}}$ and $\hat{M}_q^{\text{ref}}$, respectively. Supervising both outputs ensures stable training and facilitates gradient flow through the entire network.

To implement the proposed boundary-enhanced paradigm, we introduce explicit supervision for boundary map prediction. We employ a dedicated lightweight boundary prediction head $f_{boun}(\cdot)$, which takes intermediate features to predict a boundary map $\hat{B}_q$. The ground-truth boundary map B_q^{gt}, generated by applying the Canny edge detector to M_q, serves as the target for computing the boundary loss. To counter the severe class imbalance in boundary pixels, we leverage a weighted binary cross-entropy loss

$$L_{boun} = \text{BCEWithLogitsLoss}\left(\hat{B}_q, B_q^{gt}; pos_weight = 10\right). \tag{23}$$

A high positive weight is used to increase the recall of the sparse foreground (boundary) pixels. This loss term is critical for enforcing precise boundary delineation, directly addressing a key weakness of conventional methods, as illustrated in Fig. 1.

To mitigate the domain gap between support and query features, a feature-level domain alignment loss L_{domain} is introduced. F_s and F_q features are projected into a shared domain-invariant space using a domain projector $f_{domain}(\cdot)$, which consists of global average pooling followed by fully connected layers. The mean squared error (MSE) loss is then minimized between the projected support and query feature vectors

$$z_s = f_{\text{domain}}(F_s) \tag{24}$$

$$z_q = f_{\text{domain}}(F_q) \tag{25}$$

$$L_{domain} = \text{MSE}(z_s, z_q). \tag{26}$$

This term enforces the encoder to produce representations that are invariant to the appearance discrepancies between specific support and query instances, leading to more generalized and robust prototypes.

IV. Experiments

A. Datasets

To comprehensively evaluate the generalization ability of BS-PMN on unseen classes, we conduct experiments on three publicly available benchmarks: two RS datasets (iSAID [47] and LoveDA [48]) and one natural image dataset (PASCAL-5^i [29]).

The iSAID dataset is one of the most widely used benchmarks, which comprises 2806 images and 655 451 annotated object instances. Based on this dataset, the iSAID-5^i [1] partitioning scheme is derived, covering 15 representative semantic categories, including airplanes, ships, vehicles, buildings, and storage tanks. Following the experimental protocol proposed in [1], we adopt a threefold cross-validation strategy, in which 15 categories are divided into three disjoint subsets. In each fold, five categories are selected as the test set, and the other ten categories are used for training. The category splits for each fold are provided in Table I.

The LoveDA-2^i dataset serves as a benchmark for evaluating generalization under urban–rural domain shifts. It contains 5987 high-resolution RS images, containing 166 768 annotated semantic objects in total. The dataset covers six representative land-cover categories. We also adopt a threefold cross-validation strategy, where six categories are divided into three groups. In each fold, two categories are selected as the test set, and the other four categories are used for training. The category splits for each fold are summarized in Table I.

The PASCAL-5^i dataset consists of the PASCAL VOC 2012 [49] and additional annotations from the Semantic Boundaries Dataset (SBD) [50]. In accordance with the evaluation protocol established in base and meta (BAM) [53], we partition the 20 categories into four folds, with each fold comprising five classes for cross-validation.

B. Experimental Setup

All experiments were conducted on an NVIDIA GeForce RTX 3090 GPU and implemented in the PyTorch framework. The training process lasted for a total of 50 000 iterations. The entire model was trained in an end-to-end manner using stochastic gradient descent (SGD) with momentum. The momentum was set to 0.9, and the weight decay was fixed at 0.0005. For all experiments, we employed a batch size of 4 and set the initial learning rate to 0.002, following the poly decay strategy [51]. Input images were resized to 256 $\times$ 256 for iSAID-5^i and 473 $\times$ 473 for both LoveDA-2^i and PASCAL-5^i. During training, data augmentation techniques, such as random horizontal flipping, random scale cropping, and Gaussian blurring, were applied to enhance robustness.

For comparison, we employed ResNet-50 and VGG-16 as backbone networks, both initialized with ImageNet-pretrained parameters to enhance feature extraction and improve generalization. Following the previous work, we use the mIoU as the primary evaluation metric to evaluate segmentation performance across different classes and folds.

C. Comparison With State-of-the-Arts

1) iSAID-5^i: On the iSAID-5^i benchmark, we evaluate our approach against a broad set of state-of-the-art (SOTA) competitors. These include early prototype-based methods (PANet [31] and PFENet [52]), methods focusing on multiprototype representation, such as SDM [1] and Deep Meta-Metric Learning (DMML) [42], background suppression (BAM [53] and R^2Net [3]), knowledge distillation (PCNet [2]), self-attention aggregation (MS^2A^2Net [7]), and the very recent models, such as

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TABLE II

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE iSAID-5^i DATASET

Backbone	Method	1shot				5shot			
		Split-0	Split-1	Split-2	mIoU	Split-0	Split-1	Split-2	mIoU
VGG16	PANet (ICCV-19) [31]	26.86	14.56	20.69	20.70	30.89	16.63	24.05	23.86
	PFENet (TPAMI-20) [52]	28.52	17.05	18.94	21.50	37.59	23.22	30.45	30.42
	DMML (TGRS-22) [42]	24.41	18.58	19.46	20.82	28.97	21.02	22.78	24.26
	BAM (CVPR-22) [53]	33.93	16.88	21.47	24.09	38.46	22.76	28.81	30.01
	SDM (TGRS-22) [1]	29.24	20.80	34.73	28.26	36.33	27.98	42.39	35.57
	R^2Net (TGRS-23) [3]	35.27	19.93	24.63	26.61	**42.06**	23.52	30.06	31.88
	PCNet (TGRS-23) [2]	32.48	19.88	24.56	25.64	41.09	21.98	34.14	32.40
	MS^2A^2Net (TGRS-24) [7]	32.46	23.15	30.76	28.79	38.56	25.23	33.57	32.45
	DSMFNet (J-STARS-24) [55]	31.25	24.35	31.04	28.88	40.73	26.92	38.05	35.23
	HSE (TGRS-25) [27]	37.18	29.79	36.04	34.34	40.87	30.20	43.44	38.17
	MVLPNet (TGRS-25) [54]	**38.10**	26.72	36.42	33.75	39.57	29.74	40.89	36.73
	BS-PMN (ours)	34.25	**29.85**	**38.96**	**34.35**	39.92	**34.49**	**44.96**	**39.79**
ResNet50	PANet (ICCV-19) [31]	27.56	17.23	24.60	23.13	36.54	16.05	26.22	26.27
	PFENet (TPAMI-20) [52]	35.84	23.35	27.20	28.80	42.42	25.34	33.00	33.59
	DMML (TGRS-22) [42]	28.45	21.02	23.46	24.31	30.61	25.23	24.08	26.18
	BAM (CVPR-22) [53]	39.43	21.69	28.64	29.92	43.29	27.92	38.62	36.61
	SDM (TGRS-22) [1]	34.29	22.25	35.62	30.72	39.88	30.59	45.70	38.72
	R^2Net (TGRS-23) [3]	41.22	21.64	35.28	32.71	46.45	25.80	39.84	37.36
	PCNet (TGRS-23) [2]	40.24	24.64	31.31	32.06	45.31	28.19	37.36	36.96
	MS^2A^2Net (TGRS-24) [7]	34.96	25.30	30.71	30.32	42.30	27.10	36.31	35.24
	DSMFNet (J-STARS-24) [55]	39.03	25.92	40.57	35.17	43.69	30.81	42.32	38.94
	HSE (TGRS-25) [27]	**44.03**	**35.87**	39.26	**39.72**	47.03	36.75	45.66	43.15
	MVLPNet (TGRS-25) [54]	41.91	28.27	41.03	37.07	44.68	30.07	49.01	41.46
	BS-PMN (ours)	38.75	34.11	**41.51**	38.12	**47.32**	**38.19**	**52.73**	**46.08**

DSMFNet [55], holistic semantic embedding (HSE) [27], and MVLPNet [54].

We conduct experiments using both ResNet-50 and VGG-16 backbones under one- and five-shot settings. The quantitative results are presented in Table II. Our BS-PMN achieves highly competitive performance across most evaluation settings, especially in the five-shot scenario. With a ResNet-50 backbone, BS-PMN reaches 46.08% mIoU, outperforming recent models such as Holistic Semantic Embedding (HSE) [27] and MVLPNet [54] by +2.93% and +4.62%, respectively. Although HSE exhibits a slight lead in the one-shot ResNet-50 setting, BS-PMN demonstrates a superior capacity for knowledge integration, achieving a substantial gain of +7.33% from one-shot to five-shot. This indicates that our SPR cycle effectively leverages multiple support references to rectify prototypes and mitigate the domain gap.

A particularly noteworthy advantage is BS-PMN's exceptional performance on Split-1, which is notoriously challenging due to its dense, small, and geometrically irregular objects. While other methods exhibit severe performance degradation on this split, our framework maintains a significant lead. While methods such as R^2Net [3] and MVLPNet [54] struggle with background clutter in these scenarios, the explicit geometric constraints from our BAPE framework effectively suppress irrelevant activations. This demonstrates that our boundary-aware approach is more advanced than simple semantic-guided matching in capturing the structural essence of RS targets.

As shown in Fig. 5, our method achieves precise localization and sharp boundary delineation under both one- and five-shot settings. The predictions align closely with ground-truth masks, effectively suppressing background clutter and minimizing false positives. The explicit boundary supervision ensures geometrically coherent outputs, while the self-refinement cycle enhances internal consistency when multiple support samples are available. These qualitative observations confirm that our approach addresses key challenges in RS-FSS: accurate target localization and detailed boundary recovery.

2) LoveDA-2^i: To evaluate the generalization ability of BS-PMN on unseen classes with diverse characteristics, we conduct experiments on the LoveDA-2^i benchmark. Unlike iSAID-5^i, LoveDA-2^i focuses on land-cover classification with significant urban–rural domain shifts and irregular, texture-rich categories (e.g., agriculture, forest, and barren). As reported in Table III, BS-PMN achieves competitive or superior performance compared to the state-of-the-art methods. With a ResNet-50 backbone, our model reaches 26.25% and 30.81% mIoU under one- and five-shot settings, respectively. Notably, we incorporated R^2Net [3] as a highly competitive baseline, as it is specifically designed to handle background clutter in RS scenes. BS-PMN consistently outperforms R^2Net across most splits, validating that our framework is more robust in complex land-cover scenarios.

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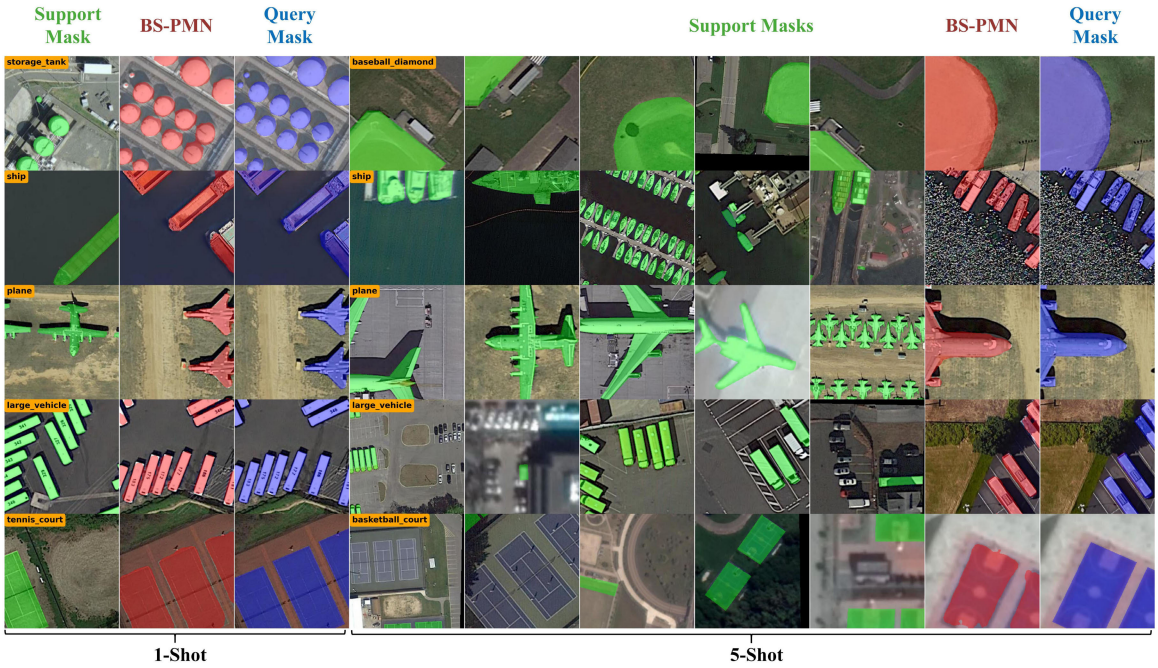


Fig. 5. Qualitative segmentation results on iSAID-5^i dataset using the ResNet-50 backbone.

TABLE III

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE LOVEDA-2^i DATASET

Backbone	Method	1shot				5shot			
		Split-0	Split-1	Split-2	mIoU	Split-0	Split-1	Split-2	mIoU
VGG16	PANet (ICCV-19) [31]	13.68	11.80	8.74	11.41	19.51	15.34	15.32	16.72
	CANet (CVPR-19) [30]	14.58	17.35	13.12	15.02	16.26	18.24	16.57	17.02
	PFENet (TPAMI-20) [52]	16.14	24.35	31.57	24.02	15.08	26.82	30.18	24.03
	ASGNet (CVPR-21) [34]	15.93	21.60	29.08	22.20	17.33	26.24	35.75	26.44
	SCL (CVPR-21) [56]	15.31	21.43	23.89	20.21	14.84	22.39	20.85	19.36
	BAM (CVPR-22) [53]	17.12	21.37	**33.68**	24.06	18.21	24.51	34.32	25.68
	DCPNet (IJCAI-22) [57]	15.22	22.58	31.83	23.21	15.94	26.38	31.37	24.56
	NTRENet (CVPR-22) [58]	15.07	23.17	28.68	22.31	15.30	25.12	30.95	23.79
	SDM (TGRS-22) [1]	12.07	21.16	34.14	22.46	21.44	31.03	36.45	29.64
	R^2Net (TGRS-23) [3]	18.58	22.18	28.83	23.20	21.75	27.86	32.51	27.37
	DMNet (TGRS-23) [4]	19.71	**26.23**	30.43	**25.46**	**25.02**	**35.51**	33.62	31.38
	BS-PMN (ours)	**21.14**	25.32	29.07	25.18	24.18	32.31	**38.42**	**31.63**
ResNet50	PANet (ICCV-19) [31]	17.99	14.42	13.25	15.22	21.67	19.26	19.68	20.20
	CANet (CVPR-19) [30]	16.83	19.33	16.32	17.49	18.01	20.41	17.58	18.67
	PFENet (TPAMI-20) [52]	17.13	22.20	26.49	21.94	15.83	25.73	24.74	22.10
	ASGNet (CVPR-21) [34]	15.91	20.21	22.33	19.48	18.38	26.29	36.34	27.00
	SCL (CVPR-21) [56]	15.14	20.45	25.00	20.20	14.25	21.09	23.65	19.66
	BAM (CVPR-22) [53]	19.35	22.26	**32.69**	24.77	22.26	25.18	31.49	26.31
	DCPNet (IJCAI-22) [57]	16.67	23.10	24.44	21.40	13.43	25.59	28.06	22.36
	NTRENet (CVPR-22) [58]	16.05	22.69	21.87	20.20	15.79	24.94	23.42	21.38
	SDM (TGRS-22) [1]	17.53	21.79	29.29	22.87	18.58	25.92	33.75	26.08
	R^2Net (TGRS-23) [3]	19.31	22.92	30.25	24.16	22.40	29.37	33.17	28.31
	DMNet (TGRS-23) [4]	19.29	**25.52**	31.53	25.45	**24.62**	**33.80**	33.12	30.51
	BS-PMN (ours)	**24.62**	23.22	30.92	**26.25**	24.13	31.45	**36.86**	**30.81**

On the LoveDA-2^i benchmark, the superiority of BS-PMN is evident in texture-rich categories, such as forest and agriculture. Unlike DMNet [4] or SDM [1], which rely on high-level semantic prototypes, our MFPM module integrates low-level structural–spectral and frequency-aware features. This allows the model to capture more discriminative prototypes that are resilient to the spectral blurring often found in these land-cover classes. Moreover, the substantial improvement from one-shot to five-shot (e.g., +6.45% with VGG-16) highlights the effectiveness of our prototype generation and fusion mech-

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TABLE IV
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE PASCAL-5^i DATASET

Method	1shot					5shot				
	Split-0	Split-1	Split-2	Split-3	mIoU	Split-0	Split-1	Split-2	Split-3	mIoU
PANet (ICCV-19) [31]	44.00	57.50	50.80	44.00	49.10	55.30	67.20	61.30	53.20	59.30
PFENet (TPAMI-20) [52]	61.70	69.50	55.40	56.30	60.80	63.10	70.70	55.80	57.90	61.90
HSNet (ICCV-21) [35]	64.30	70.70	60.30	60.50	64.00	70.30	73.20	67.40	67.10	69.50
VAT (ECCV-22) [59]	67.60	72.00	62.30	60.10	65.30	72.40	73.60	68.60	65.70	70.10
BAM (CVPR-22) [53]	68.97	73.59	67.55	61.13	67.81	70.59	75.05	70.79	67.20	70.91
DMNet (TGRS-23) [4]	64.55	72.52	61.50	57.82	64.10	67.39	74.01	64.93	62.08	67.10
SDM (TGRS-22) [1]	68.84	72.11	65.51	71.41	69.47	70.68	74.68	66.01	72.15	70.88
HDMNet (CVPR-23) [60]	71.00	75.40	68.90	62.10	69.40	71.30	76.20	71.30	68.50	71.80
RPMG (TCSVT-23) [61]	64.40	72.60	57.90	58.40	63.30	65.30	72.80	58.40	59.80	64.10
DRNet (TCSVT-24) [62]	66.10	68.80	61.30	58.20	63.60	69.20	73.90	65.40	65.30	68.50
DCP (IJCV-24) [63]	67.20	72.90	65.20	59.40	66.10	70.50	75.30	68.00	67.70	70.30
MSANet (OJCS-25) [64]	69.25	74.60	67.84	62.40	68.52	72.70	76.26	73.52	67.94	72.60
SPANet (IAENG-25) [65]	70.30	**76.50**	**68.98**	62.77	69.64	72.29	**76.89**	**74.70**	67.81	72.92
PRFomer (TCSVT-25) [66]	70.20	75.00	67.30	65.40	69.50	72.40	76.80	70.40	68.30	71.90
BS-PMN (ours)	**72.70**	73.65	66.12	**74.74**	**71.80**	**76.08**	74.91	67.20	**75.31**	**73.38**

anism in leveraging multiple support samples, offering a clear advantage over competing methods.

Fig. 6 presents a qualitative comparison of five-shot segmentation results between our BS-PMN and two representative methods (SDM [1] and R^2Net [3]) across the iSAID-5^i and LoveDA-2^i benchmarks. On the iSAID-5^i dataset, SDM and R^2Net frequently suffer from boundary blurring or incomplete coverage; in contrast, BS-PMN produces segmentation masks with significantly sharper boundaries and higher structural integrity. Specifically, for categories such as ships and storage tanks, our method achieves precise localization and successfully delineates intricate object contours where competing methods exhibit crude or misaligned results. This confirms the efficacy of the BAPE module in preserving fine-grained spatial details and enforcing geometric constraints. On the LoveDA-2^i dataset, which is characterized by complex textures and spectral similarity, our model demonstrates superior robustness. Compared to the noisy or fragmented predictions of SDM and R^2Net, BS-PMN provides markedly more precise boundaries. More importantly, by preventing the masks from leaking into spectrally similar background areas, our model successfully avoids oversegmentation in challenging categories, such as water, barren land, and forest. This visual evidence validates that the proposed SPR mechanism effectively mitigates intraclass variance and spectral blurring, allowing our model to generalize better across diverse RS scenarios.

The consistently high-quality results on both datasets, despite their vastly different characteristics, provide strong qualitative evidence that our framework learns a more generalized and powerful representation for few-shot segmentation, effectively overcoming the distinct challenges posed by both instance-centric and land-cover-oriented tasks.

3) PASCAL-5^i: To further evaluate the cross-domain generalization capability of BS-PMN, we conducted extensive evaluations on the PASCAL-5^i natural image benchmark, as summarized in Table IV. Our framework achieves average mIoU scores of 71.80% and 73.38% under the one- and five-shot settings, respectively, consistently outperforming recent state-of-the-art models, such as SPANet [65] and PRFormer [66]. Notably, BS-PMN exhibits a remarkable competitive edge in Split-3. While Split-3 presents significant challenges due to heavy occlusions and the intricate, high-frequency boundaries of categories such as potted plants, sofas, and tv/monitors, our model achieves a peak one-shot performance of 74.74%. This outperforms the representative multiprototype method, SDM [1] (71.41%). This leap underscores the efficacy of the BAPE framework in providing precise boundary localization and the MFPM module's robustness in capturing diverse object appearances within cluttered indoor backgrounds. In the five-shot scenario, the performance further scales to 73.38% with high stability across all categories, from transportation to living beings, empirically validating the SPR cycle's capacity for effective domain adaptation. Overall, these results confirm that BS-PMN possesses high versatility and generalization value beyond Earth observation tasks.

D. Ablation Study

1) Ablation Studies of Main Modules: To validate the effectiveness of each key component in the proposed BS-PMN, we conduct extensive ablation experiments on the iSAID-5^i dataset in the five-shot setting. The experimental results are systematically reported in Table V. The baseline model is implemented with a ResNet-50 backbone, achieving an mIoU of 41.23%. Subsequently, we incrementally integrate the MFPM module, the BAPE framework, and the SPR cycle to scrutinize their individual and collective contributions.

1) The MFPM module serves as the foundational feature extraction and prototype generation component. As

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TABLE V
ABLATION STUDY ON THE iSAID-5^i DATASET

MFPM	BAPE	SPR	Split-0	Split-1	Split-2	mIoU
			40.88	36.17	46.65	41.23
✓			45.93	36.11	52.05	44.69
	✓		46.56	35.70	48.83	43.69
		✓	42.01	36.69	50.52	43.07
✓	✓		46.19	**38.31**	51.15	45.21
✓		✓	45.97	37.78	52.17	45.31
	✓	✓	44.40	37.17	50.86	44.15
✓	✓	✓	**47.32**	38.19	**52.73**	**46.08**

shown in the second row of Table V, equipping the baseline with the complete MFPM module, which integrates frequency-aware decomposition, structural–spectral similarity, and EM clustering, brings a substantial performance gain of +3.46% in mIoU. This significant improvement underscores the critical importance of leveraging multisource and frequency-aware features for generating discriminative prototypes in complex RS scenes. It effectively mitigates the issue of overlooking crucial low-level cues, thereby providing a more robust representation for subsequent matching.

2) The BAPE framework, which incorporates explicit boundary supervision and boundary-aware prototype fusion, is designed to address the challenge of blurred boundaries. When added alone to the baseline (the third row of Table V), it improves the mIoU by +2.46% (to 43.69%). More notably, its synergy with the MFPM module is powerful, yielding a combined improvement of +3.98% (to 45.21%) over the baseline. This result demonstrates that BAPE successfully augments the semantically rich prototypes from MFPM with precise geometric priors. The explicit boundary loss enforces accurate delineation, directly tackling the high boundary error rates of conventional methods. The performance gain is consistent across splits, particularly in Split-1, which is known for its challenging objects, proving the module's robustness.
3) The SPR cycle aims to alleviate the support–query domain gap by bootstrapping the model's own predictions. Independently, it improves performance by +1.84%. When combined with the MFPM module, the system achieves a top mIoU of 45.31% among all two-component combinations. This strong result confirms that the initial prototypes from MFPM are of high quality, enabling the self-learning cycle to produce accurate pseudolabels. These labels, in turn, generate more target-aware and balanced query prototypes. This process shifts the model's approach from "learning from others" to "self-reliance," which is crucial for generalizing to unseen query instances.
4) The full model, which integrates all three proposed components (MFPM, BAPE, and SPR), achieves a state-of-the-art performance of 46.08% mIoU. This represents a remarkable +4.85% absolute improvement over a strong baseline. The key to this superior performance lies in our hierarchical and synergistic architecture, where each module is designed to address a specific challenge: feature representation, boundary ambiguity, and the domain gap. Comprehensive ablation studies conclusively validate the necessity and effectiveness of each component.

TABLE VI
ABLATION STUDY ON THE INTERNAL COMPONENTS OF THE MFPM MODULE

FDF	SSC	Split-0	Split-1	Split-2	mIoU
✓		47.34	37.07	52.61	45.67
	✓	**47.36**	37.09	50.42	44.96
✓	✓	47.32	**38.19**	**52.73**	**46.08**

TABLE VII
ABLATION STUDY ON THE INTERNAL COMPONENTS OF THE BAF MODULE

Fusion Method	Split-0	Split-1	Split-2	mIoU
Sum	44.95	37.09	47.80	43.28
BAF (w/o Gate Fusion)	**48.34**	37.50	51.78	45.87
BAF	47.32	**38.19**	**52.73**	**46.08**

The overall performance gain (+4.85% mIoU) is not merely a stack of modules, but a result of our hierarchical and synergistic architecture. Each module has a clearly defined academic boundary: MFPM purifies features, BAPE rectifies geometries, and SPR adapts domains. This targeted design directly addresses the fundamental contradictions in RS few-shot segmentation that generic SOTA methods fail to resolve.

2) Ablation Studies of Key Components in MFPM: We evaluate the internal design of the MFPM module by ablating its two key elements: the frequency-aware decomposition and fusion (FDF) module and the structural–spectral similarity channel (SSC) computation. The results are presented in Table VI.

The experimental results provide clear evidence that both components are indispensable for building robust and discriminative prototypes, while contributing in distinct yet complementary ways. The FDF module improves stability (45.67% mIoU) by suppressing noise, whereas the SSC module enhances discriminability (44.96% mIoU) by leveraging low-level geometric cues. Their synergistic combination achieves the best performance (46.08% mIoU), effectively tackling intraclass variance and interclass similarity.

3) Ablation Studies of Key Components in BAF: To validate the internal design of the BAF module, we conduct an ablation study comparing different strategies for integrating the foreground and boundary prototypes. The results are systematically reported in Table VII.

The results demonstrate a clear and progressive improvement in performance, supporting each of our architectural decisions. The most basic elementwise sum (Sum) fusion yields the lowest mIoU (43.28%), indicating that treating

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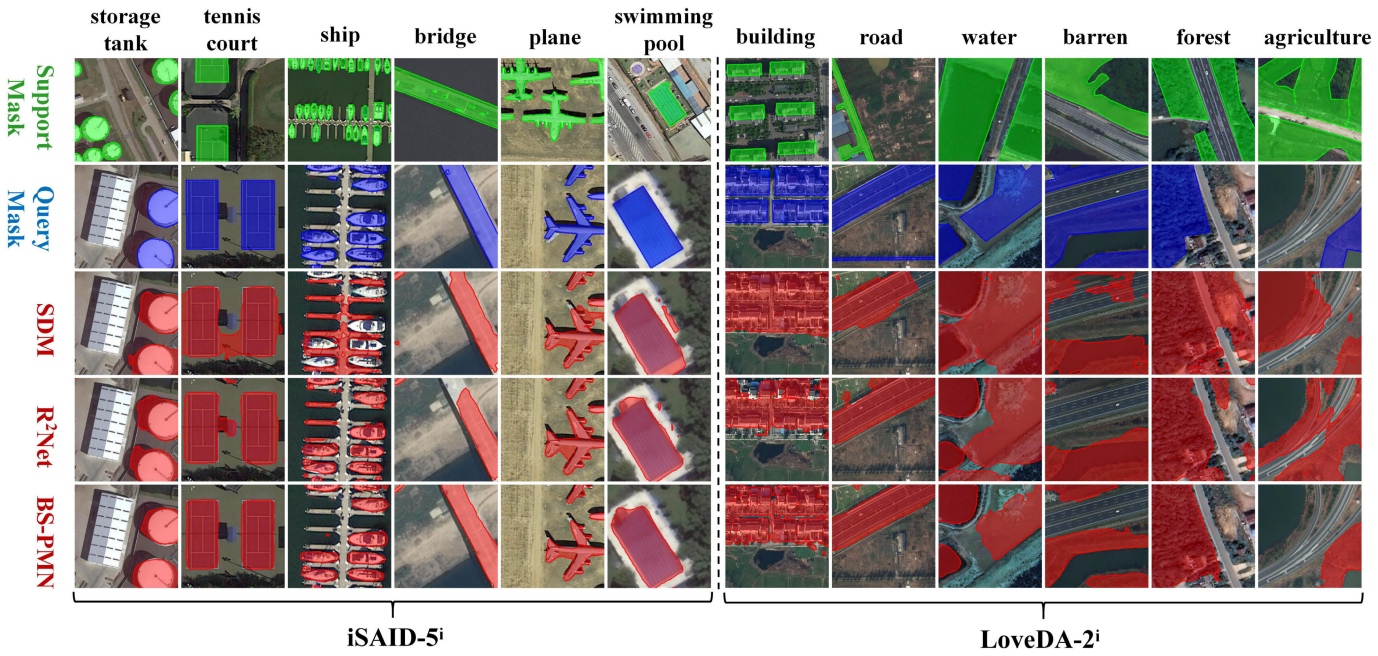


Fig. 6. Qualitative comparison of five-shot segmentation results on both iSAID-5^i and LoveDA-2^i datasets.

semantic and boundary information as equally important is ineffective. This naive addition fails to model the complex interplay between object features and boundary cues, resulting in poor localization. Replacing it with our cross-attention mechanism (BAF w/o gate fusion) brings a significant gain of +2.59% mIoU. This improvement confirms that modeling the relevance between foreground and boundary prototypes is critical. Cross-attention enables the model to dynamically weight boundary information for each semantic concept, producing geometrically informed semantic prototypes.

The incorporation of the final gating mechanism (Full BAF) raises the performance to the peak of 46.08% mIoU. The gain over the nongated version, while modest, is consistent and achieves the top score on the most challenging Split-1. This indicates that the gating mechanism provides a crucial refinement. It acts as an adaptive filter that fine-tunes the cross-attention output, controlling information flow to preserve the most useful elements from both the original semantic input and the refined output. This process prevents oversmoothing and noise, thereby enhancing the model's robustness.

4) Qualitative Analysis: To intuitively demonstrate the effectiveness of the proposed BAPE framework and SPR cycle, we provide a visual comparison of segmentation results (see Fig. 7) alongside quantitative ablation experiments on the iSAID-5i dataset (see Table V). The qualitative results align closely with the numerical gains, corroborating the complementary roles of each component.

Comparing Fig. 7(b) and (c) reveals the profound impact of explicit boundary modeling. The baseline predictions, while capturing the general semantic area, exhibit blurred contours and a tendency to activate irrelevant background regions. In contrast, the predictions from boundary-enhanced prototypes demonstrate dramatically sharper and more accurate boundaries. This visual evidence directly validates the core function of the BAPE framework: it injects crucial geometric priors into the semantic prototypes, enforcing precise delineation and effectively suppressing background clutter, as intended by our design.

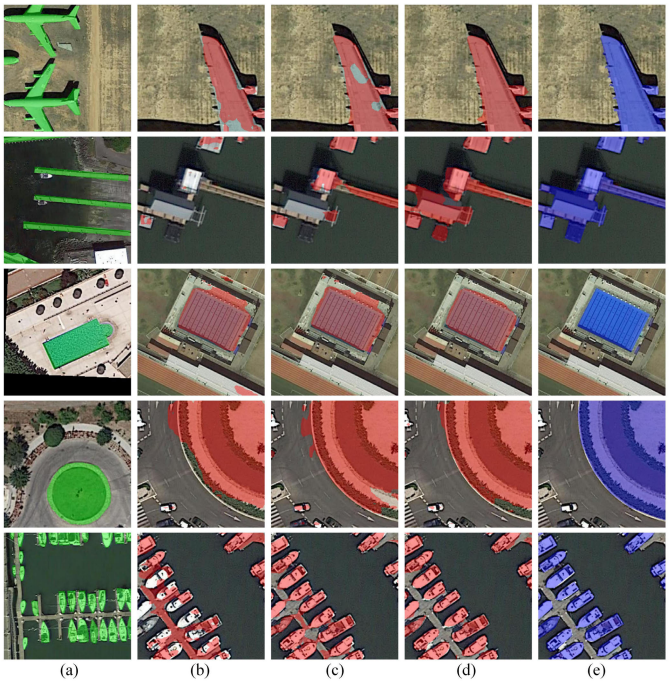


Fig. 7. Qualitative results comparing different prototype configurations. From left to right: (a) support image (mask). (b) Prediction using baseline prototypes. (c) Prediction using boundary-enhanced prototypes (MFPM+BAPE). (d) Prediction using self-refined prototypes (MFPM+BAPE+SPR). (e) Query image (mask).

The comparison between Fig. 7(c) and (d) highlights the additional benefit of the SPR cycle. While the boundary-

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enhanced model defines boundaries well, it may occasionally undersegment complex objects or miss parts of them due to the residual support–query domain gap. The full model, which bootstraps query-specific prototypes through self-learning, produces more complete and internally consistent segmentations. For instance, in the first and second rows, the full model more accurately captures the entire extent of the object without the minor omissions present in Fig. 7(c). This demonstrates that the SPR cycle successfully adapts the general, support-derived knowledge to the specific context of the query image, generating more target-aware prototypes and alleviating the domain gap.

In addition to the segmentation masks, we visualize the class activation maps (heatmaps) to intuitively illustrate the model's focus area throughout the refinement process. Fig. 8 presents a comparative analysis of heatmaps generated by three configurations, confirming the quantitative and qualitative findings from earlier.

The heatmaps from the baseline model [see Fig. 8(b)], which generates prototypes by directly applying the EM algorithm to support features, reveal its core weakness: a lack of precise spatial awareness. The high-activation (red) regions exhibit a low congruence with the ground-truth masks. As observed in samples such as the ship (row 1) and the airplane (row 3), the model's focus is dispersed, often activating irrelevant background areas or missing key parts of the foreground. This indicates that although the EM-clustered prototypes capture certain semantic information, the absence of multifeature fusion and frequency-aware mechanisms results in biased prototype representations, which, in turn, lead to imprecise and noisy matching with query features. This visual evidence clearly explains the underlying cause of blurred boundaries and false activations observed in the baseline predictions.

The introduction of the BAPE framework, building upon the prototypes generated by the MFPM module, brings a remarkable improvement in spatial precision [see Fig. 8(c)]. The heatmaps show a significantly higher alignment with the ground-truth masks. The high-activation regions become more concentrated and accurately overlaid on the target objects. This demonstrates that the boundary-aware prototypes provide a strong geometric prior, effectively guiding the model to concentrate its attention on the correct foreground regions while suppressing activation in distracting background areas. The enhanced focus is a direct visual correlate of the sharper boundaries achieved by this component.

Our complete model yields the most accurate and reliable heatmaps [see Fig. 8(d)]. The responses are not only precise but also complete and internally consistent. In contrast to the dispersed focus of the baseline [see Fig. 8(b)], the full model achieves a homogeneous activation across the entire object with near-zero spillover into the background. This optimal performance stems from the synergistic interaction of all three proposed components: the MFPM module first generates robust and discriminative prototypes through multifeature and frequency-aware clustering; the BAPE module then refines them with explicit boundary constraints to ensure fine-grained spatial alignment; finally, the SPR cycle further adapts these prototypes to the query image, mitigating the support–query domain gap and enabling the generation of target-specific prototypes that homogeneously activate all relevant parts of the object.

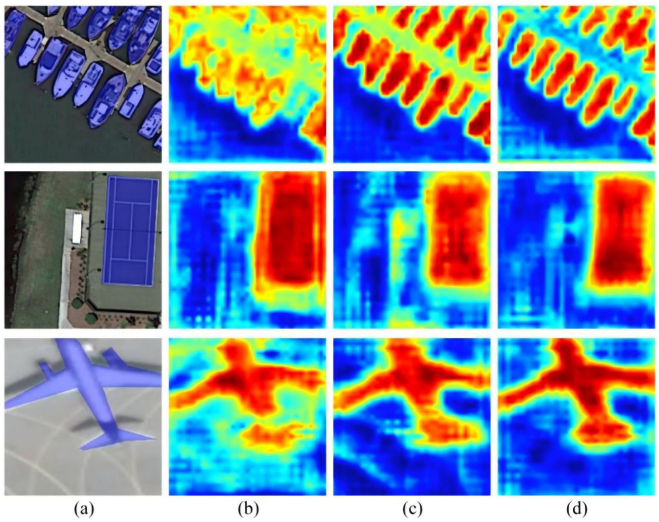


Fig. 8. Comparison of heatmaps of pixelwise similarity between the encoded features and the prototypes. (a) Query image (mask). The subsequent columns display heatmaps from (b) baseline, (c) model with BAPE (MFPM+BAPE), and (d) our full model (MFPM+BAPE+SPR). Red regions indicate the highest activation, signifying the model's focus.

To further validate the quality of the prototypes generated by different configurations, we project the high-dimensional prototype vectors onto a 2-D space using t-SNE [67]. This visualization provides an intuitive understanding of the discriminative power and compactness of the learned representations. As illustrated in Fig. 9, we observe a clear progressive improvement of prototypes in the organization of the feature space.

a) Original Prototypes (Baseline): The prototypes of different classes (e.g., blue versus orange points) show significant overlap and dispersion. This indicates that the baseline features, while containing semantic information, lack sufficient discriminativity. The blurred boundaries in the feature space directly explain the blurred segmentation boundaries and misclassifications observed in Fig. 7 and the dispersed activation in Fig. 8.

b) Boundary-Aware Prototypes (MFPM + BAPE): With the incorporation of boundary-aware enhancement, the prototypes become more clustered within the same class, and the margin between different classes increases. The geometric constraints provided by the BAPE framework help to purify the prototypes, pulling them away from background-related features and making them more focused on the core attributes of the target. This leads to the sharper attention and improved accuracy seen in the qualitative results.

c) Self-Refined Prototypes (MFPM + BAPE + SPR): This further employs self-learning refinement and produces the most well-organized feature space. The prototypes form tight, distinct clusters for each class with maximal separation between them. This demonstrates that the SPR cycle successfully alleviates the support–query domain gap, generating

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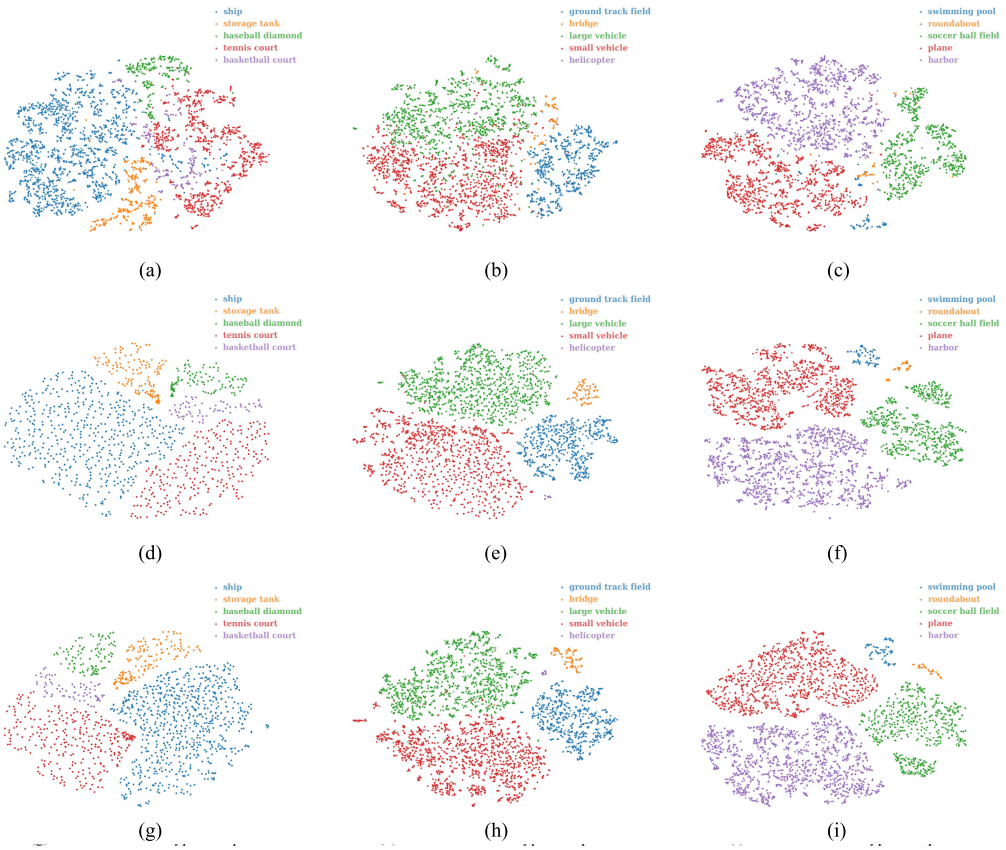


Fig. 9. t-SNE visualization of different prototypes. (a) Original prototypes of split-0. (b) Original prototypes of split-1. (c) Original prototypes of split-2. (d) Boundary-aware prototypes of split-0. (e) Boundary-aware prototypes of split-1. (f) Boundary-aware prototypes of split-2. (g) Self-refined prototypes of split-0. (h) Self-refined prototypes of split-1. (i) Self-refined prototypes of split-2.

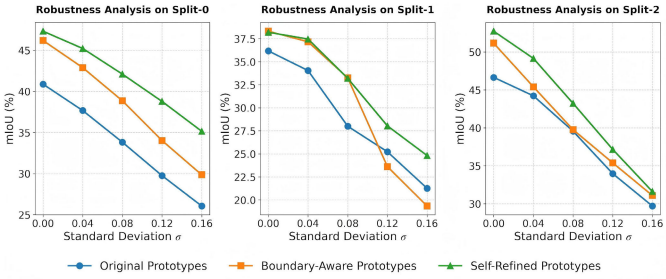


Fig. 10. Robustness analysis of BS-PMN against stochastic disturbances.

highly target-aware and balanced prototypes that are optimal for few-shot matching. This superior clustering property is the fundamental reason behind our model's state-of-the-art performance.

E. Model Analysis

1) Hyperparameter Sensitivity:

a) Loss Function Coefficients: An ablation study on the coefficients of the key loss terms, the boundary loss L_{boun} and the domain alignment loss L_{domain}, was conducted to validate their necessity and our chosen weighting. The segmentation loss coefficients were fixed at the common setting of $\lambda_{init} = 0.4$ and $\lambda_{ref} = 0.6$.

The most severe performance degradation occurs when either loss is completely deactivated. As shown in Table VIII, removing the boundary supervision ($\lambda_{boun} = 0$) causes a substantial performance drop of up to −1.13 mIoU. This sharp decline highlights the critical importance of our explicit boundary constraint for achieving high segmentation accuracy. Similarly, removing the domain alignment loss ($\lambda_{domain} = 0$) also causes a significant decrease up to −1.10 mIoU, confirming that mitigating the support–query domain gap is essential for learning robust and generalizable prototypes. Furthermore, adjusting the coefficients around our default configuration ($\lambda_{boun} = 0.2$ and $\lambda_{domain} = 1.0$) leads to only minor performance variations. Compared to completely removing these loss terms, perturbations within this range result in only slight performance degradation. This robustness suggests that the benefits of both loss terms are consistently delivered once

TABLE VIII
ABLATION STUDY ON THE COEFFICIENTS OF LOSS FUNCTIONS

λ_{boun}	λ_{domain}	mIoU	Δ mIoU
0.2	1.0	**46.08**	-
0.0	1.0	44.95	-1.13
0.5	1.0	45.87	-0.21
0.2	0.0	45.32	-0.76
0.2	0.5	45.79	-0.29
0.0	0.5	44.23	-1.85
0.0	0.0	43.51	-2.57
0.5	0.5	45.35	-0.73
0.5	0.0	44.98	-1.10

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TABLE IX
ABLATION STUDY ON THE NUMBER OF PROTOTYPES

K	mIoU	Δ mIoU
1	43.12	-2.96
3	45.45	-0.63
5	**46.08**	-
7	45.82	-0.26
10	45.67	-0.41

they are activated, and the model does not overfit to these hyperparameter values.

b) Number of Prototypes: We further investigate the impact of the number of prototypes K generated by the MFPM module, which is a critical parameter for capturing the high intraclass variance in RS imagery. As summarized in Table IX, we evaluated $K \in \{1, 3, 5, 7, 10\}$. A single prototype ($K = 1$) results in the lowest performance (43.12% mIoU), confirming that a single feature vector is insufficient to represent the complex spectral and textural variations of RS geo-objects. The performance significantly improves as K increases, reaching a peak at $K = 5$. Beyond this point ($K = 7, 10$), the gain in mIoU saturates and even slightly decreases due to the inclusion of redundant noise and increased computational complexity. This empirical evidence justifies our choice of $K = 5$ as the optimal balance between representational capacity and model efficiency.

2) Robustness to Noise: To evaluate the generalization ability of BS-PMN in complex RS scenarios, we conducted a robustness analysis by simulating stochastic disturbances and sensor-related artifacts (e.g., atmospheric scattering or electronic noise). Specifically, we injected varying levels of Gaussian noise with standard deviation $\sigma \in \{0.00, 0.04, 0.08, 0.12, 0.16\}$ into the query images during inference.

As visualized in Fig. 10, the performance of all models inevitably degrades as noise intensity increases; however, our full BS-PMN (self-refined prototypes) exhibits remarkable resilience compared to the baseline and the intermediate MFPM+BAPE variant. Notably, the performance curve for BS-PMN is significantly flatter, particularly in Split-0 and Split-1. Although the performance drop on Split-2 appears relatively more pronounced because geometry-dependent categories such as planes and roundabouts rely heavily on sharp structural features that are easily obscured by Gaussian noise, BS-PMN, nevertheless, maintains a consistent mIoU margin over competing methods across all noise levels.

This stability is attributed to the synergistic interaction of our core modules: the MFPM ensures robust initial feature clustering, the BAPE framework imposes explicit geometric constraints that act as stable anchors, and the SPR cycle effectively mitigates noise-induced error propagation by iteratively refining query-aware prototypes. These results provide strong empirical evidence that BS-PMN is capable of maintaining reliable segmentation under environmental interference, making it suitable for deployment in low-quality imaging conditions.

3) Model Complexity and Efficiency: To demonstrate the practical feasibility of BS-PMN, we compare its number of parameters (Para.) and FLOPs with several state-of-the-art methods in Table X. All metrics are measured with a ResNet-50 backbone and a 256×256 input size. Compared to existing methods, BS-PMN achieves an excellent tradeoff between accuracy and complexity. Notably, BS-PMN maintains the lowest parameter count (18.9M) among all compared models, yet it delivers a significantly higher mIoU than the recent MVLPNet while requiring fewer FLOPs (97.2 versus 120.4G).

TABLE X
COMPARISON OF MODEL COMPLEXITY ON THE iSAID-5^i DATASET

Method	mIoU	Params	FLOPs
PANet (ICCV-19) [31]	26.27	23.6M	599.7G
PFENet (TPAMI-20) [52]	33.59	53.4M	156.3G
BAM (CVPR-22) [53]	36.61	51.6M	174.1G
SDM (TGRS-22) [1]	38.72	19.6M	**91.55G**
R^2Net (TGRS-23) [3]	37.36	51.8M	339.6G
MVLPNet (TGRS-25) [54]	41.46	205.1M	120.4G
BS-PMN (ours)	**46.08**	**18.9M**	97.2G

Furthermore, our model is substantially more efficient than BAM and R^2Net in terms of both model size and computational cost. Specifically, compared to R^2Net, BS-PMN reduces the FLOPs by 71.3% (97.2 versus 339.6G) while providing a massive performance gain of +8.72% mIoU. These results confirm that our architectural innovations, particularly the targeted boundary and self-learning modules, provide high-precision RS segmentation without imposing an excessive computational burden.

V. CONCLUSION

In this article, we address the pivotal challenges of blurred boundaries and high intraclass variance in few-shot segmentation of RS images. To this end, we have proposed BS-PMN. Our framework integrates three key components: first, the MFPM module exploits multisource features, including spatial-based frequency-decomposed components and structural–spectral cues, to generate highly discriminative prototypes; second, the BAPE framework incorporates geometric boundaries to augment semantic prototypes, effectively resolving ambiguity in object outlines; and finally, an SPR cycle is designed to close the support–query domain gap, iteratively refining target-aware prototypes. Evaluations on benchmarks confirm that our approach achieves state-of-the-art results. BS-PMN delivers superior performance, excelling particularly in generating accurately bounded segmentation masks, as evidenced by both quantitative and qualitative analyses.

Despite its effectiveness, BS-PMN possesses inherent limitations that offer avenues for future research. First, the boundary supervision currently relies on heuristic edge extraction, which may degrade in complex textures; thus, incorporating learnable contour modeling could further improve reliability. Second, the self-learning process remains sensitive to initial prediction quality, suggesting that uncertainty-aware optimization or adaptive confidence thresholding could

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enhance stability. Finally, extending our boundary-aware paradigm to multimodal data, such as Synthetic Aperture Radar (SAR) or LiDAR, represents a promising direction for achieving superior performance in heterogeneous environments. We believe that these advancements will pave the way toward more precise and label-efficient scene understanding in RS.

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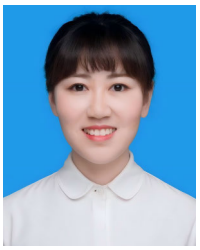
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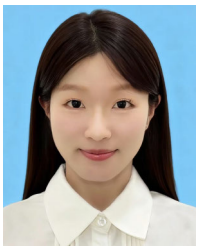
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